PROBABILITY AND STATISTICS

CASE STUDY

ON

STUDY OF RELATION BETWEEN ROAD ACCIDENTS AND BAD WEATHER CONDITIONS

BY

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Question: Do the bad weather conditions increase the no. of road accidents?

- **Collect a sample of road accidents occurred due to heavy rain fall. (both quantity of rainfall and the number of accidents is to be collected)**
- **Find the coefficient of correlation between the rain fall and accidents.**
- **Construct a quadratic regression curve suitable for the data collected.**

Here is the collection of scores of a sample of students in 'Introduction to Programming' course both theoretical and practical exams.

SERIAL NUMBERS	No. of Accidents	Temperature of the day
1	2	30
2	3	27
3	5	24
4	1	25
5	4	22
6	2	22
7	1	23
8	4	24
9	2	25
10	1	23

$$\bar{X}=[2+3+5+1+4+2+1+4+2+1]/10$$
$$=25/10$$
$$=2.5$$

$\bar{y}$=[30+27+24+25+22+22+23+24+25+23]/10

=245/10

=24.5

CORELLATION COEFFICIENT BETWEEN ROAD ACCIDENTS & WEATHER CONDITIONS

Let x be road accidents and y be the bad weather conditions

x	X= $(x-\bar{x})$	X^2	y	Y= $(y-\bar{y})$	Y^2	XY
1	**-0.5**	**0.25**	**30**	**5.5**	**30.25**	**-2.75**
2	**0.5**	**0.25**	**27**	2.2	6.25	1.25
3	2.5	6.25	24	-0.5	0.25	**-1.25**
4	-1.5	2.25	25	0.5	**0.25**	**-0.75**
5	1.5	2.25	22	-2.5	6.25	3.75
6	-0.5	0.25	22	-2.5	6.25	1.25
7	-1.5	2.25	23	-1.5	2.25	**2.25**
8	1.5	2.25	24	-0.5	0.25	**-0.75**
9	-0.5	0.25	25	1.5	0.25	**-0.75**

10	-1.5	2.25	23	-1.5	0.25	0.25

CORRELATION COEFFICIENT:

The degree of relationship between the variables under consideration is measured through the correlation analysis.

The extended degree of relationship between the two variables is measured in terms of another parameter called coefficient of correlation.

Coefficient of correlation is denoted by "r".

The value of r lies between -1 and 1, i.e., $-1 \quad r \quad 1$. From the above table

from the above bable

ΣX = 25

ΣY = 245

AX = 2.5

Y = 24.5

ΣXY = -2.5

ΣX^2 = 18-5

ΣY^2 = 54.5

Correlation Coefficient r = $\dfrac{\Sigma XY}{\sqrt{\Sigma X^2 * \Sigma Y^2}}$

$$= \frac{-2.5}{\sqrt{(18.5)*(54.5)}}$$

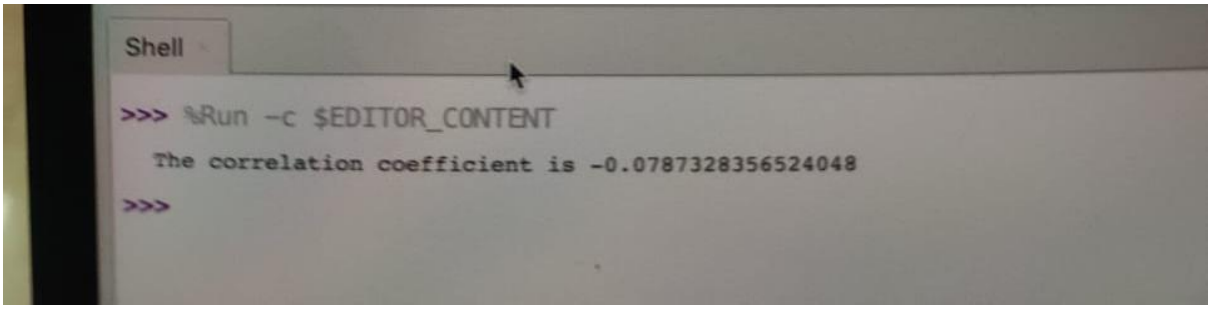
$$= -0.248975$$

Hence the correlation coefficient between the road accidents and bad weather conditions is

CODE:

```
import numpy as np
data1=np.array([2,3,5,1,4,2,1,4,2,1])
data2=np.array([30,27,24,25,22,22,23,24,
25,23])
x_bar=np.mean(data1)
y_bar=np.mean(data2)
nr=sum((data1-x_bar) *(data2-y_bar))
dr=np.sqrt(sum((data1-
x_bar)**2)*sum((data2-y_bar)**2))
corr=nr/dr
print("The correlation coefficient is",corr)
```

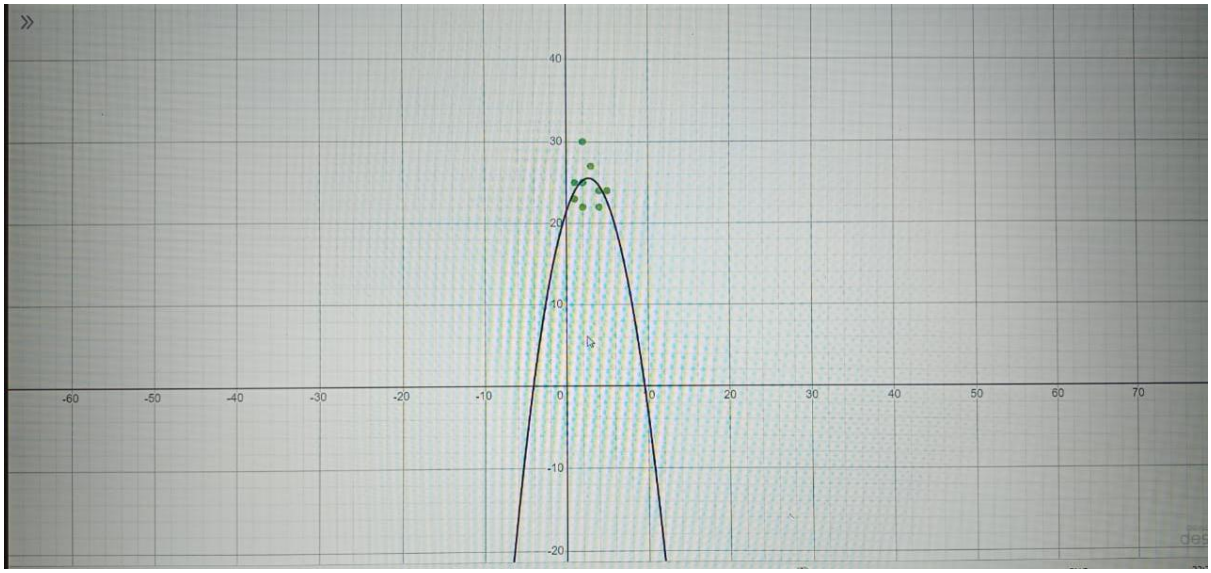
OUTPUT:



Construct a quadratic regression curve suitable for the data collected.

X1=2,3,5,1,4,2,1,4,2,1

Y1=30,27,24,25,22,22,23,24,25,23



Y1=ax1^2+bx1+c

a=-0.547323

b=2.9565

c=21.5421

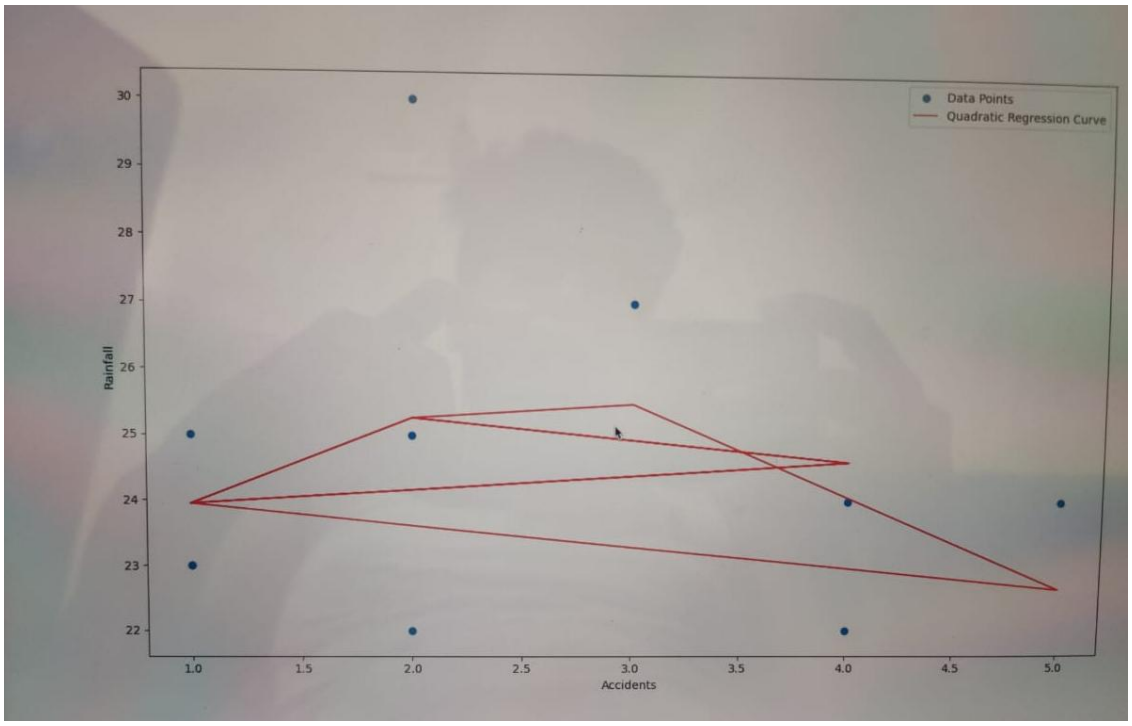
Y1=-0.5473x1^2+2.9565x1+21.5421

Code:

```
import numpy as np
import matplotlib.pyplot as plt
from numpy.polynomial import Polynomial
from scipy.stats import linregress
rainfall = np.array([your rainfall data])
accidents = np.array([your accidents data])
coefficients = np.polyfit(rainfall, accidents, 2)
quadratic_curve = np.poly1d(coefficients)
print("Quadratic Regression Equation:")
print(quadratic_curve)
```

```
plt.scatter(rainfall, accidents,
label='Data Points')
plt.plot(rainfall,
quadratic_curve(rainfall), color='red',
label='Quadratic Regression Curve')
plt.xlabel('Rainfall')
plt.ylabel('Accidents')
plt.legend()
plt.show()
```

output:



REFERENCE:

These are the references which are collected from the random hospitals

18	15-6-2023	Sandeep	Knee injury	19	M
19	15-6-2023	Kiran	food Poison	39	M
20	15-6-2023	Jyothi	Accident	43	f
21	15-6-2023	Rajesh	mouth Disease	21	M
21	15-6-2023	Arvind	Accident	20	M
22	15-6-2023	Aisha	Accident	9	f

18	15-6-2023	Sandeep	Knee injury	19	M
19	15-6-2023	Kiran	food Poison	39	M
20	15-6-2023	Jyothi	Accident	43	f
21	15-6-2023	Rajesh	mouth Disease	21	M
21	15-6-2023	Arvind	Accident	20	M
22	15-6-2023	Aisha	Accident	9	f

3	29-6-2023	Neha	leg pain	19	f
4	29-6-2023	Ananya	Accident	20	f
5	29-6-2023	Swetha	Accident	33	f
6	29-6-2023	Mahesh	Accident	37	M
7	29-6-2023	Suresh	fever	26	M
8	29-6-2023	Radhika	Accident	28	f
9	29-6-2023	Vinay	Accident	18	M
10	29-6-2023	Pooja	Delivery	21	f

19	5-8-2023	Varsha	Delivery	24	f
20	5-8-2023	Arushi	Accident	13	f
21	5-8-2023	Rafath	Accident	22	M
22	5-8-2023	Vihaan	stomach ache	7	M
23	5-8-2023	Radhika	Accident	47	f
24	5-8-2023	Neha	Typhoid	19	f
25	5-8-2023	lakshman	Accident	15	M

AccuWeather Hyderabad, Telangana 25°c

28	29	30	31	1	2	3
39° 27°	40° 29°	34° 26°	39° 25°	41° 26°	41° 28°	42° 30°
4	5	6	7	8	9	10
38° 25°	38° 25°	40° 27°	40° 29°	40° 27°	40° 27°	39° 27°
11	12	13	14	15	16	17
38° [illegible]°	39° 27°	39° 27°	39° 27°	39° 27°	39° 27°	39° 27°
18	19	20	21	22	23	24
39° 28°	38° 29°	39° 29°	39° 24°	36° 24°	33° 26°	32° 23°
25	26	27	28	29	30	1

AccuWeather Hyderabad, Telangana 25°c Search

July 2023 DAILY →

S	M	T	W	T	F	S
25	26	27	28	29	30	1
31° 23°	31° 25°	32° 24°	33° 24°	33° 24°	34° 24°	35° 25°
2	3	4	5	6	7	8
34° 25°	34° 24°	34° 25°	32° 23°	31° 23°	32° 24°	32° 24°
9	10	11	12	13	14	15
33° 24°	33° 24°	29° 23°	29° 22°	32° 24°	34° 24°	30° 25°
16	17	18	19	20	21	22